$$\mathbf{m=0:V_0} = \{\{0,0,0,0,0,0,0,0,0\},\{0,0,0,0,0,0,0,0,0\},\{0,0,0,0,0,0,0,0,0\},\{0,0,0,0,0,0,0,0,0\},\{0,0,0,0,0,0,0,0,0\},\{0,0,0,0,0,0,0,0,0\},\{0,0,0,0,0,0,0,0,0\},\{0,0,0,0,0,0,0,0,0\},\{0,0,0,0,0,0,0,0,0\}\}$$

$$\mathbf{m=2:P_1} = \begin{pmatrix} 2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 \end{pmatrix}, \mathbf{m=2:P_2} = \begin{pmatrix} 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \end{pmatrix},$$

$$\mathbf{m=2:P_3} = \begin{pmatrix} 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & \omega & \omega^2 & 0 & 0 \\ 0 & 0 & 0 & \omega^2 & 0 & 0 & 0 & \omega & 0 \\ 0 & 0 & \omega & 0 & 0 & 0 & 0 & \omega^2 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & \omega^2 & 0 & 0 & 0 & 0 & \omega & 0 & 0 \\ 0 & \omega & 0 & 0 & 0 & \omega^2 & 0 & 0 & 0 \\ 0 & 0 & \omega^2 & \omega & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \end{pmatrix}, \mathbf{m=2:P_4} = \begin{pmatrix} 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & \omega^2 & \omega & 0 & 0 \\ 0 & 0 & 0 & \omega & 0 & 0 & 0 & \omega^2 & 0 \\ 0 & 0 & \omega^2 & 0 & 0 & 0 & 0 & \omega & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & \omega & 0 & 0 & 0 & 0 & \omega^2 & 0 & 0 \\ 0 & \omega^2 & 0 & 0 & 0 & \omega & 0 & 0 & 0 \\ 0 & 0 & \omega & \omega^2 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \end{pmatrix},$$

$$\mathbf{m=4:W_12} = \begin{pmatrix} 2 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & -1 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & -1 & 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & -1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 & -1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 0 & -1 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 2 \end{pmatrix}, \mathbf{m=4:W_13} = \begin{pmatrix} 2 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & -1 & 0 & 0 & 0 & \omega & \omega^2 & 0 & 0 \\ 0 & 0 & -1 & \omega^2 & 0 & 0 & 0 & \omega & 0 \\ 0 & 0 & \omega & -1 & 0 & 0 & 0 & \omega^2 & 0 \\ 1 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 1 \\ 0 & \omega^2 & 0 & 0 & 0 & -1 & \omega & 0 & 0 \\ 0 & \omega & 0 & 0 & 0 & \omega^2 & -1 & 0 & 0 \\ 0 & 0 & \omega^2 & \omega & 0 & 0 & 0 & -1 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 2 \end{pmatrix},$$

$$\mathbf{m=4:W_14} = \begin{pmatrix} 2 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & -1 & 0 & 0 & 0 & \omega^2 & \omega & 0 & 0 \\ 0 & 0 & -1 & \omega & 0 & 0 & 0 & \omega^2 & 0 \\ 0 & 0 & \omega^2 & -1 & 0 & 0 & 0 & \omega & 0 \\ 1 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 1 \\ 0 & \omega & 0 & 0 & 0 & -1 & \omega^2 & 0 & 0 \\ 0 & \omega^2 & 0 & 0 & 0 & \omega & -1 & 0 & 0 \\ 0 & 0 & \omega & \omega^2 & 0 & 0 & 0 & -1 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 2 \end{pmatrix}, \mathbf{m=4:W_23} = \begin{pmatrix} 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 & 0 & -\omega^2 & -\omega & 0 & 0 \\ 0 & 0 & 0 & -\omega & 0 & 0 & 0 & -\omega^2 & 0 \\ 0 & 0 & -\omega^2 & 0 & 0 & 0 & 0 & -\omega & 0 \\ 2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 \\ 0 & -\omega & 0 & 0 & 0 & 0 & -\omega^2 & 0 & 0 \\ 0 & -\omega^2 & 0 & 0 & 0 & -\omega & 0 & 0 & 0 \\ 0 & 0 & -\omega & -\omega^2 & 0 & 0 & 0 & 0 & 0 \\ 2 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 \end{pmatrix},$$

$$\mathbf{m=4:W_24} = \begin{pmatrix} 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 & 0 & -\omega & -\omega^2 & 0 & 0 \\ 0 & 0 & 0 & -\omega^2 & 0 & 0 & 0 & -\omega & 0 \\ 0 & 0 & -\omega & 0 & 0 & 0 & 0 & -\omega^2 & 0 \\ 2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 \\ 0 & -\omega^2 & 0 & 0 & 0 & 0 & -\omega & 0 & 0 \\ 0 & -\omega & 0 & 0 & 0 & -\omega^2 & 0 & 0 & 0 \\ 0 & 0 & -\omega^2 & -\omega & 0 & 0 & 0 & 0 & 0 \\ 2 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 \end{pmatrix},$$

$$\mathbf{m=4:W_34} = \begin{pmatrix} 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 & 0 & -1 & -1 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & 0 & -1 & 0 \\ 2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 \\ 0 & -1 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & -1 & -1 & 0 & 0 & 0 & 0 & 0 \\ 2 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 \end{pmatrix}, \mathbf{m=6:U_1} = \begin{pmatrix} 0 & 0 & 0 & 0 & 3 & 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 3 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 3 & 0 & 0 & 0 & 3 & 0 & 0 & 0 & 0 \end{pmatrix},$$

$$\mathbf{m=6:U_2} = \begin{pmatrix} 2 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 2 \\ 0 & -1 & 0 & 0 & 0 & -1 & -1 & 0 & 0 \\ 0 & 0 & -1 & -1 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & -1 & -1 & 0 & 0 & 0 & -1 & 0 \\ 2 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 2 \\ 0 & -1 & 0 & 0 & 0 & -1 & -1 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & -1 & -1 & 0 & 0 \\ 0 & 0 & -1 & -1 & 0 & 0 & 0 & -1 & 0 \\ 2 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 2 \end{pmatrix}, \mathbf{m=6:U_3} = \begin{pmatrix} 2 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 2 \\ 0 & -1 & 0 & 0 & 0 & -\omega & -\omega^2 & 0 & 0 \\ 0 & 0 & -1 & -\omega^2 & 0 & 0 & 0 & -\omega & 0 \\ 0 & 0 & -\omega & -1 & 0 & 0 & 0 & -\omega^2 & 0 \\ 2 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 2 \\ 0 & -\omega^2 & 0 & 0 & 0 & -1 & -\omega & 0 & 0 \\ 0 & -\omega & 0 & 0 & 0 & -\omega^2 & -1 & 0 & 0 \\ 0 & 0 & -\omega^2 & -\omega & 0 & 0 & 0 & -1 & 0 \\ 2 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 2 \end{pmatrix},$$

$$\mathbf{m=6:U_4} = \begin{pmatrix} 2 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 2 \\ 0 & -1 & 0 & 0 & 0 & -\omega^2 & -\omega & 0 & 0 \\ 0 & 0 & -1 & -\omega & 0 & 0 & 0 & -\omega^2 & 0 \\ 0 & 0 & -\omega^2 & -1 & 0 & 0 & 0 & -\omega & 0 \\ 2 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 2 \\ 0 & -\omega & 0 & 0 & 0 & -1 & -\omega^2 & 0 & 0 \\ 0 & -\omega^2 & 0 & 0 & 0 & -\omega & -1 & 0 & 0 \\ 0 & 0 & -\omega & -\omega^2 & 0 & 0 & 0 & -1 & 0 \\ 2 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 2 \end{pmatrix}, \mathbf{m=8:sl_3} = \begin{pmatrix} 2 & 0 & 0 & 0 & 3 & 0 & 0 & 0 & 3 \\ 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ 3 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ 3 & 0 & 0 & 0 & 3 & 0 & 0 & 0 & 2 \end{pmatrix}\}$$